

- **scikit-learn:** Implements machine learning algorithms for predictive modelling
- **Zipline/Backtrader:** Provides frameworks for backtesting trading strategies

These libraries allow traders to build sophisticated systems without "reinventing the wheel." For example, calculating a simple moving average (a common trading indicator) can be done in just a few lines:

```
1. # A very simple moving average calculation
2. stock_prices = [150, 152, 154, 158, 160] # Last 5 days of prices
3.
4. # Calculate a 3-day moving average (average of last 3 prices)
5. day_1_to_3_average = (stock_prices[0] + stock_prices[1] + stock_prices[2]) / 3
6. day_2_to_4_average = (stock_prices[1] + stock_prices[2] + stock_prices[3]) / 3
7. day_3_to_5_average = (stock_prices[2] + stock_prices[3] + stock_prices[4]) / 3
8.
9. print("Moving averages:", day_1_to_3_average, day_2_to_4_average, day_3_to_5_average)
10.
```

With libraries like pandas, this becomes even easier:

```
1. # Using pandas to calculate moving averages
2. import pandas as pd
3.
4. # Create a series of stock prices
5. prices = pd.Series([150, 152, 154, 158, 160])
6.
7. # Calculate a 3-day moving average
8. moving_average = prices.rolling(window=3).mean()
9.
10. print("Moving averages:", moving_average)
11.
```